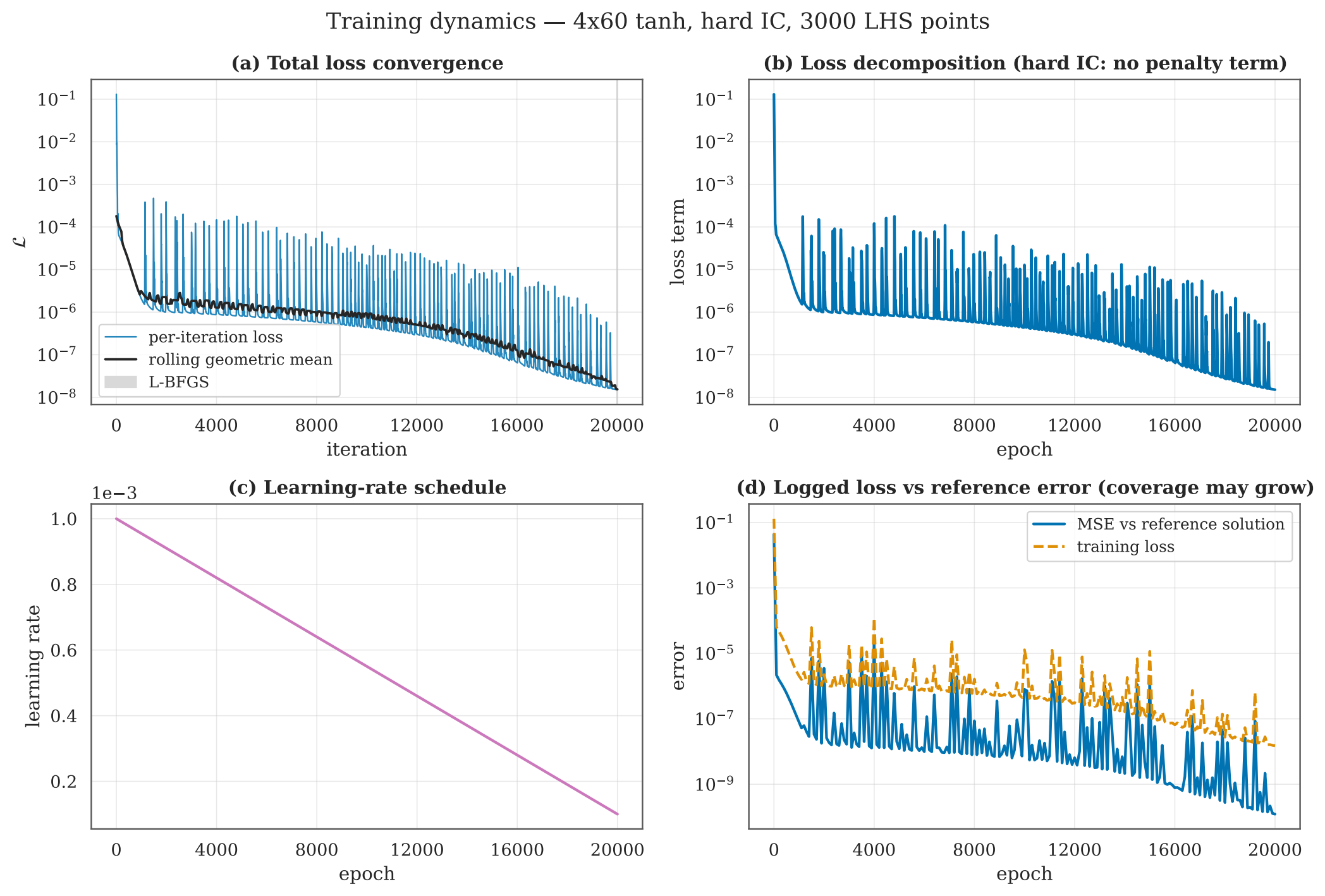

Training dynamics — 4x60 tanh, hard IC, 3000 LHS points
(a) Total loss convergence
$\mathcal{L}$
per-iteration loss
rolling geometric mean
L-BFGS
iteration
(b) Loss decomposition (hard IC: no penalty term)
loss term
epoch
(c) Learning-rate schedule
1e−3
learning rate
epoch
(d) Logged loss vs reference error (coverage may grow)
MSE vs reference solution
training loss
error
epoch